

Varen	6869	AgSp1.1	serine	rm2	1	PGTTPGTVITGRDCKPK.KFIVPNGAFTTPCSFPGPDGEPISVEPAGPGCTTPGIEITDSDGNIKKLVVPTITKGPGGYPMGPGSPFG	84
Varen	6869	AgSp1.2	serine	rm2	1	PGTTPGVVITGPDGQ.PVKYIVPREAFSTPCSTPGPHGKRVHVHPAGPGCTTPGAKTSDSDGVKSIVLNTTPKE.S....GPGF...	76
Varen	6869	AgSp1.1	serine	rm2	2	PGTTPGTVITGRDCKPK.KFIVPNGAFTTPCSFPGPDGEPISVEPAGPGCTTPGIEITDSDGNIKKLVVPTIT	69
Varen	6869	AgSp1.2	serine	rm2	2	PGTTPGVVITGPDGQPVKYIVPREAFSTPCSTPGPHGKRVHVHPAGPGCTTPGAKTSDSDGVKSIVLNTTPKE.S....GPGF...	77
Varen	6869	AgSp1.2	serine	rm2	3	PGTTWGVVITGPDGQ.....PVKGDGQPIQIVPAGPGCTTLGCVITGPDGQPVKPRAVITEPDGEP.....	57
Msag	19329	AgSp1.1	serine	rm2	1	PGITPGAKTSDSDGVESIVVPSTPKGS.S.....	28
Msag	19329	AgSp1.1	serine	rm2	16	YVVPQGAFTTPCSFAQVHCKRVHVQFAG.....	28
Msag	19329	AgSp1.1	serine	rm2	17	PGITPGAKTSDSDGVESIVVPSTPKGSD.....	28
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	2	PGTTPGVVITGPDGQPV.KFIVPQGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGAKTNSDGSIDSIVLPTTPK.....	70
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	3	PGTTPGVVITGPDGQPVK.....	17
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	6		17
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	7	PGTTPGVVITGPDGQPVK.....	17
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	17		0
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	18	PGTTPGVVITGPDGQPV.KFIVPQGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGAKTNSDGSIDSIVLPTTPKG.S....GPAFPTT	79
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	19	PGTTPGVVITGPDGQPVK.....	17
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	22		0
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	23	PGTTPGVVITGSDGKPV.KFIVPQGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGAKIDSNGSIDSIVL.....PTT	68
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	24	PGTTPGVVITGPDGQPV.KFIVPQGAFTTPCSIPGPHGKPIHVHPAGPGCTTPGAKIDSNGSIDSIVLPTTPKG.S....GPAF...	76
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	25	PGTTPGVVITGPDGQPV.KFIVPNGAFSTPCSTPGPNKGPIHVHPAGPGCTTPGAKLSDSDGDISEIFLPSTLSRPAPQP.....	76
Atri	AgSp1	genomic	serine	rm2	1	PGTTPGTVITGPDCKPN.KFVLPKGAFTTPCSIPGPDGKPIHVHPAGPGCTTPGAKLSDSDGDISEIFLPSTLSRPAPQP.....	84
Atri	AgSp1	genomic	serine	rm2	2	PCSTPGTITGPDCKPKK.....	17
Atri	AgSp1	genomic	serine	rm2	41	PGTTPGTVITGPDCKPN.KFVLPKGAFTTPCSIPGPDGKPIHVHPAGPGCTTPGTVITGPDGSIINQVVFVPTTPKSSEASGGKKGTGEN	84
Atri	AgSp1	genomic	serine	rm2	42	PCSTPGTITGPDCKPK.KFVVPQGAFTTPCSVPGPDGNPIVVEPAGPGCTTPGTQTGPDGKINKVVPVPTTPNPKGPESQPGNPLS	84
Atri	AgSp1	genomic	serine	rm2	43	PGTTPGTVITGRNGRPK.RFVVPKGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGVITGSDGSIDSIVLPTTPKES.....	72
Atri	AgSp1	genomic	serine	rm2	44	PGTTPGVVITGPDGKPV.KFVVPFGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGAETSDSDGIDSIVLPSTP.....	69
Atri	AgSp1	genomic	serine	rm2	45	PGTTPGVVITGPDGQPV.EFIVPFGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGAKTSDSDGIDSIVLPSTP.....	69
Aarg	AgSp1	FC511	serine	rm2	1	PGTTPGTVITGRNGRPK.RFVVPKGAFTTPCSIPGPHGKPIHVHPAGPGCTTPGVKTGPDGSIIDSIVLPTTPKES....QPGF...	76
Aarg	AgSp1	FC511	serine	rm2	3b	PGTTPGVVITGPDGRPV.KFVVPQGAFTTPCSFTGPNKGPIHVHPAGPGCTTPGAKTNSDGSIDSIVLPTTPKG.S....GPGF...	76
Aarg	AgSp1	FC511	serine	rm2	2	PGTTPGVVITGPDGQPF.KFIVPFGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGAKTSDSDGIDSIVLPSTPKG.P....SPGL...	76
Aarg	AgSp1	FC511	serine	rm2	3	PGTTPGVVITGPDGCEPI.EFIVPFGAFTTPCSIPGPNKGPIHVHPAGPGCTTPGAKTSDSDGIDSIVLPSTPKG.P....SPGF...	76
Aarg	AgSp1	FC511	serine	rm2	4	PGTTPGVVITGPDGRPV.KFIVPQGAFTTPCSFMGPNKGPIHVHPAGPGCTTPGAKTSDSDGIDSIVLPRTPRG.A....GPGF...	76
Aarg	AgSp1	FC511	serine	rm2	5	VGTTPGIVITGPDGQPV.KFIVPNGAFSTTCYITPGPHGKPIHVHPAGPGCTTPGAKTSDSDGVSEIILPSTPSPPAPK.....	75
Mhut	1271	AgSp1	serine	rm2	1	PGATPGAKTSDSGSVESIVLPATPFGSG.....	28
Mhut	1271	AgSp1	serine	rm2	2	PGTTPGVITNRDGQPV.EFIVPQGALRTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFR.T....G....	73
Mhut	1271	AgSp1	serine	rm2	3	PGTTPGVITNRDGQPV.EFIVPQGAFTTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFG.S....G....	73
Mhut	1271	AgSp1	serine	rm2	4	PGTTPGVITNRDGQPV.EFIVPQGALRTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFG.T....G....	73
Mhut	1271	AgSp1	serine	rm2	5	PGTTPGVITNRDGQPV.EFIVPQGAFTTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFG.S....G....	73
Mhut	1271	AgSp1	serine	rm2	6	PGTTPGVITNRDGQPV.EFIVPQGALRTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFR.T....G....	73
Mhut	1271	AgSp1	serine	rm2	7	PGTTPGVITNRDGQPV.EFIVPQGAFTTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFG.S....G....	73
Mhut	1271	AgSp1	serine	rm2	8	PGTTPGVITNRDGQPV.EFIVPQGALRTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFR.T....G....	73
Mhut	1271	AgSp1	serine	rm2	9	PGTTPGVITNRDGQPV.EFIVPQGAFTTPGTIKGPHGKPIHVHPAGPGATPGAKTSDSGSVESIVLPATPFG.S....G....	73
Mhut	1271	AgSp1					

Varen 6869 AgSp1.1 serine rm2 1	PGNMFQPMR.....P.GRTTTPSSVEGPDGEPIQIEPAG	117
Varen 6869 AgSp1.2 serine rm2 1	D.FHTPSHVTKGDGKPIQIVPAG	98
Varen 6869 AgSp1.1 serine rm2 2		69
Varen 6869 AgSp1.2 serine rm2 2	D.FHTPSHVTKGDGKPIQIVPAG	99
Varen 6869 AgSp1.2 serine rm2 3	IQIIPAG	64
Msag 19329 AgSp1.1 serine rm2 1	PGFQSPTPRRVTKPACQPIETVAAGPGTTPGVVTGPDGQPVRVVPQGAFITTPGSFAGLH	88
Msag 19329 AgSp1.1 serine rm2 16		28
Msag 19329 AgSp1.1 serine rm2 17	PGFRAPTPSRATEFTGQPIQIVTTG	53
Lcorn 51676 AgSp1.1Cfrag serine rm2 2	..G.....SGPGFQFQTPCPITKPGGQPIQIVPAG	98
Lcorn 51676 AgSp1.1Cfrag serine rm2 3		17
Lcorn 51676 AgSp1.1Cfrag serine rm2 6	GSGPGIQFQTPCPITKPGGQPIQIVPAG	28
Lcorn 51676 AgSp1.1Cfrag serine rm2 7		17
Lcorn 51676 AgSp1.1Cfrag serine rm2 17	GAGPGFQFQTPCPITKPGGQPIQIVPAG	28
Lcorn 51676 AgSp1.1Cfrag serine rm2 18	PRG.....SGPGFQFQTPCPITKPGGQPIQIVPAG	109
Lcorn 51676 AgSp1.1Cfrag serine rm2 19		17
Lcorn 51676 AgSp1.1Cfrag serine rm2 22	GSGPGFQFQTPCPVTPPGGQPIQIVLAG	28
Lcorn 51676 AgSp1.1Cfrag serine rm2 23	PKG.....SGPGFQFQTPCAVTKPSGQPIQIVPAG	98
Lcorn 51676 AgSp1.1Cfrag serine rm2 24	QFPAPA.PRQSGGQPIQIVPAG	97
Lcorn 51676 AgSp1.1Cfrag serine rm2 25	VNPTTPTDTRGPQGNPILIVPAG	99
Atri AgSp1 genomic serine rm2 1	PANT.....NNAIGPLQPG.SATTPTSISGPDGQPERIVPAG	120
Atri AgSp1 genomic serine rm2 2		17
Atri AgSp1 genomic serine rm2 41	PANS.....NNALGPLQPA.STTTTPTSISGPDGCPVQVVPAG	120
Atri AgSp1 genomic serine rm2 42	PNG.....QTTPGSEGPGFNFQTPRPPIKGPCKGKPIQIIPAG	120
Atri AgSp1 genomic serine rm2 43	QPGFPTYQFQTPKPIKGPCKGKRIQVVPAG	101
Atri AgSp1 genomic serine rm2 44	KGPSPGQLQFQTPRPPIKGPLCKPLQIIPAF	98
Atri AgSp1 genomic serine rm2 45	KGPSPGQLQFQTPRSVTKPDGQPIQVVPAG	98
Aarg AgSp1 FC511 serine rm2 1	PTF.....QFQTPRPPIKGPCKGKRIQVVPAG	101
Aarg AgSp1 FC511 serine rm2 3b	QFQTPKSVTKPDGQPIQVVPAG	98
Aarg AgSp1 FC511 serine rm2 2	QFQTPKPIKGPACGKPIQIIPAG	98
Aarg AgSp1 FC511 serine rm2 3	QFQTPRPITAPGCKPIQVVPAG	98
Aarg AgSp1 FC511 serine rm2 4	QFSTPGVPQPDGQPIQIVPAG	98
Aarg AgSp1 FC511 serine rm2 5	PANPTTPTDVKGPQGGPVMIIIPAG	99
Mhut 1271 AgSp1 serine rm2 1	PGSGFQTTEPITKPDGEPIHVVPAG	53
Mhut 1271 AgSp1 serine rm2 2	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 3	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 4	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 5	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 6	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 7	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 8	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 9	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 10	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 11	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 12	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 13	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 14	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 15	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 16	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 17	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 18	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 19	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 20	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 21	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 22	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 23	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 24	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 25	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 26	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 27	PGSGFQTTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 28	PGSGFMTEPITKPDGEPIHVVPAG	98
Mhut 1271 AgSp1 serine rm2 29	PGSGFQTTEPITKPDGEPIHVVPAG	98

Varen	6869	AgSp1.1	serine	rm2	1	..	117
Varen	6869	AgSp1.2	serine	rm2	1	..	98
Varen	6869	AgSp1.1	serine	rm2	2	..	69
Varen	6869	AgSp1.2	serine	rm2	2	..	99
Varen	6869	AgSp1.2	serine	rm2	3	..	64
Msag	19329	AgSp1.1	serine	rm2	1	GK	90
Msag	19329	AgSp1.1	serine	rm2	16	..	28
Msag	19329	AgSp1.1	serine	rm2	17	..	53
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	2	..	98
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	3	..	17
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	6	..	28
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	7	..	17
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	17	..	28
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	18	..	109
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	19	..	17
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	22	..	28
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	23	..	98
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	24	..	97
Lcorn	51676	AgSp1.1Cfrag	serine	rm2	25	..	99
Atri	AgSp1	genomic	serine	rm2	1	..	120
Atri	AgSp1	genomic	serine	rm2	2	..	17
Atri	AgSp1	genomic	serine	rm2	41	..	120
Atri	AgSp1	genomic	serine	rm2	42	..	120
Atri	AgSp1	genomic	serine	rm2	43	..	101
Atri	AgSp1	genomic	serine	rm2	44	..	98
Atri	AgSp1	genomic	serine	rm2	45	..	98
Aarg	AgSp1	FC511	serine	rm2	1	..	101
Aarg	AgSp1	FC511	serine	rm2	3b	..	98
Aarg	AgSp1	FC511	serine	rm2	2	..	98
Aarg	AgSp1	FC511	serine	rm2	3	..	98
Aarg	AgSp1	FC511	serine	rm2	4	..	98
Aarg	AgSp1	FC511	serine	rm2	5	..	99
Mhut	1271	AgSp1	serine	rm2	1	..	53
Mhut	1271	AgSp1	serine	rm2	2	..	98
Mhut	1271	AgSp1	serine	rm2	3	..	98
Mhut	1271	AgSp1	serine	rm2	4	..	98
Mhut	1271	AgSp1	serine	rm2	5	..	98
Mhut	1271	AgSp1	serine	rm2	6	..	98
Mhut	1271	AgSp1	serine	rm2	7	..	98
Mhut	1271	AgSp1	serine	rm2	8	..	98
Mhut	1271	AgSp1	serine	rm2	9	..	98
Mhut	1271	AgSp1	serine	rm2	10	..	98
Mhut	1271	AgSp1	serine	rm2	11	..	98
Mhut	1271	AgSp1	serine	rm2	12	..	98
Mhut	1271	AgSp1	serine	rm2	13	..	98
Mhut	1271	AgSp1	serine	rm2	14	..	98
Mhut	1271	AgSp1	serine	rm2	15	..	98
Mhut	1271	AgSp1	serine	rm2	16	..	98
Mhut	1271	AgSp1	serine	rm2	17	..	98
Mhut	1271	AgSp1	serine	rm2	18	..	98
Mhut	1271	AgSp1	serine	rm2	19	..	98
Mhut	1271	AgSp1	serine	rm2	20	..	98
Mhut	1271	AgSp1	serine	rm2	21	..	98
Mhut	1271	AgSp1	serine	rm2	22	..	98
Mhut	1271	AgSp1	serine	rm2	23	..	98
Mhut	1271	AgSp1	serine	rm2	24	..	98
Mhut	1271	AgSp1	serine	rm2	25	..	98
Mhut	1271	AgSp1	serine	rm2	26	..	98
Mhut	1271	AgSp1	serine	rm2	27	..	98
Mhut	1271	AgSp1	serine	rm2	28	..	98
Mhut	1271	AgSp1	serine	rm2	29	..	98

 non-conserved
 similar
 $\geq 50\%$ conserved